Design and Development of a Skills Gap Analysis Tool for ICT TVET Graduates in Kicukiro District, Rwanda

BSc. in Software Engineering

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Link to template

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DECLARATION

I, Fred Niyonshuti, hereby state that I have not presented this research proposal “Design and Development of a Skills Gap Analysis Tool for ICT TVET Graduates in Kicukiro District, Rwanda” anywhere else for academic or publication purposes.

The information that has been gathered from other authors, researchers, journals, books and online sources has been appropriately referenced and acknowledged in line with academic standards and APA referencing guidelines.

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LIST OF ACRONYMS/ABBREVIATIONS

Acronym	Meaning
API	Application Programming Interface
CBET	Competency-Based Education and Training
CSS	Cascading Style Sheets
ERD	Entity Relationship Diagram
GDP	Gross Domestic Product
HTML	HyperText Markup Language
ICT	Information and Communication Technology
JSON	JavaScript Object Notation
LMS	Learning Management System
NST1	National Strategy for Transformation

RTB	Rwanda TVET Board
RISA	Rwanda Information Society Authority
SQL	Structured Query Language
TVET	Technical and Vocational Education and Training
UML	Unified Modelling Language
UI	User Interface
UX	User Experience
VS Code	Visual Studio Code
REST	Representational State Transfer

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

In the context of Rwanda's development agenda, Technical and Vocational Education and Training (TVET) is a key strategy for creating a skilled workforce, while the National Strategy for Transformation (NST1, Vision 2050) explicitly targets to "develop Rwandans into capable and skilled people" through enhanced education (TVET Competency-Based Curriculum Development Framework, 2023). The strategic plans of the Ministry of Education (MoE) support this vision by focusing on competency-based TVET curricula and assessments (TVET Competency Based Curriculum Development Framework, 2023). Rwanda TVET Board (RTB), according to this model, is tasked with setting up and ensuring competency-based curricula and occupational standards for TVET levels 1-5 (TVET Competency Based Curriculum Development Framework, 2023). The competency-based approach ensures that training programmes are well connected to the needs of the labour market and international best practices.

The Information and Communication Technology (ICT) industry has become a major economic and employment-driving sector in Rwanda. In Rwanda, the government is implementing an ICT strategy to make the country an East African digital hub with significant investments in infrastructure and human capital (ICT Skills Snapshot, 2022) (Newfarmer & Twum, 2022). By 2020, the ICT sector accounted for approximately 2% of GDP and directly employed approximately 9,000 individuals in Rwanda (ICT Skills Snapshot, 2022). The National ICT Strategy (2019-2024) clearly emphasises job creation and ICT skills development, with a vision to develop a “highly skilled and knowledgeable labour force” through the provision of VIET (Newfarmer & Twum, 2022). These initiatives highlight the essential role ICT skills play in Rwanda's development; graduates of TVET programmes should be able to meet industry standards of competencies in programming, networking, cybersecurity and related skills.

In spite of policy promises, there is a large skills gap in Rwanda's TVET system. Employability is a challenge faced by many TVET graduates, including those from ICT-related courses, due to the mismatch between their competencies and industry standards (World Bank Group , 2025; Hakizayezu & Maniraho, 2022). According to the World Bank Group (2025), before recent changes, a majority of Rwanda's youth and graduates had difficulties in being employable because their skills did not align with employers' expectations. National labour surveys reflect this: overall, youth unemployment in 2023 was approximately 20.8% of the youth, and 21.7% of youth who completed TVET programmes (Rwanda State of Skills Supply and Demand Report 2024, 2024). Similarly, only 40% of TVET school graduates were employed after training as opposed to 59% of graduates of Rwanda Polytechnic (Tracer study on the employment status of TVET school and Rwanda Polytechnic graduates, 2021). Qualitative evaluations confirm these quantitative results: Rwandan TVET stakeholders describe problems with the availability of hands-on training and curricula that do not adequately target the needs of the workplace (Hakizayezu & Maniraho, 2022). All of this shows that there is a significant discrepancy between the skills that the RTB standards aim to deliver and the skills possessed by graduates.

1.2 Problem statement

One of the major difficulties is the lack of systematic instruments used for evaluating TVET learners' skills against the national standards. In Rwanda, competency assessment is usually based on informal feedback from the instructor or at the end of the course, and does not provide learners with a clear understanding of their strengths and weaknesses (World Bank Group , 2025; Hakizayezu & Maniraho, 2022). In the fast-evolving ICT industry, this translates to graduates not knowing which skills they should develop to enhance their capabilities. If not, both learners and educators cannot be certain about the exact skill gaps in relation to the ICT occupational standards of the RTB. This helps to continue to perpetuate competency gaps – when graduates join the workforce, they are not sure what skills they have and what skills they need to acquire as described by the RTB.

An automated, software-based solution is definitely needed here. A Skills Gap Analysis Tool could be used to assess the skills of each ICT graduate through practical tests, projects and/or self-reports and then compare them to the official ICT competency standards established by the Rwanda TVET Board. The system would, by quantifying the gap between the competencies achieved by the learners and what was expected, give feedback to the learners, trainers and institutions. To this end, it would inform students about their skill areas that need improvement, recommend curriculum modifications, and, in the end, promote the match between TVET outcomes and the industry expectations (World Bank Group, 2025)(Hakizayezu & Maniraho, 2022).

1.3 Project Main Objective

The main objective of this study is to design and develop a web-based Skills Gap Analysis Tool that analyses ICT TVET graduates' competencies against Rwanda TVET Board (RTB) occupational standards in order to identify skills gaps and improve employability alignment within Kicukiro District, Rwanda.

1.3.1 Specific Objectives

1. To assess the competency levels of at least 50 ICT TVET graduates in Kicukiro District across selected RTB occupational competencies using practical assessments, portfolio evidence, academic records, and self-assessment data.

2. To design and develop a web-based Skills Gap Analysis Tool that maps graduates' competencies against RTB occupational standards and generates automated recommendations for skills improvement.
3. To test and evaluate the effectiveness, usability, and accuracy of the developed system using a sample of 50–120 ICT TVET graduates within a three-month implementation period.
4. To evaluate whether the developed Skills Gap Analysis Tool improves competency awareness among ICT TVET graduates by measuring changes in competency-awareness scores using pre-assessment and post-assessment surveys.

1.4 Research questions

This study is guided by the following research questions derived from the identified problem of skills mismatch and lack of structured competency assessment among ICT TVET graduates in Kicukiro District:

1. What challenges do ICT TVET graduates in Kicukiro District face in aligning their competencies with RTB occupational standards?
2. What limitations exist in the current methods used to assess ICT competencies among TVET graduates?
3. What functional and non-functional requirements are needed to develop an effective Skills Gap Analysis Tool?

4. How effective is the proposed Skills Gap Analysis Tool in identifying competency gaps and improving employability alignment?

1.5 Project scope

Given the available time, resources and technical capacity, and the need to achieve meaningful and reliable results, the scope of this research is limited.

The study will focus on Kicukiro District of Kigali City in Rwanda geographically. This is an area with many Technical Vocational Education and Training (TVET) institutions providing programmes in ICT. The small population of Kicukiro District provides a manageable population size, yet the area is very representative and relevant for the study on TVET education and employment in urban areas in Rwanda.

The study will only target ICT TVET graduates in Kicukiro district. These are graduates who have undertaken training and are moving into the labour market, and are therefore the most suitable group to determine whether their skills are a gap to occupational standards. A sample of about 50-120 ICT TVET graduates will be used for testing the system, which is a realistic sample for the pilot implementation phase. The sample size is adequate to test system performance, usability and competency gaps determination but does not pose an excessive development and testing burden.

The project will be implemented as a web-based application to be used by small and medium-scale users. The system will enable main functions like the registration of graduates, upload of acquired competencies, mapping of skills to the Rwanda TVET Board (RTB) occupational standards and automated skill gap reports. The system does not need to be designed for nationwide deployment or large-scale institutional integration initially and will be optimised for concurrent usage within the defined pilot range.

In terms of scope, this is only regarding competencies related to ICT, which results in a focused and in-depth review of the skills within one domain. At this stage, the system will not cover other TVET disciplines like construction, hospitality and agriculture. This restriction is required to maintain the accuracy for competency mapping and to be able to assess the prototype in the chosen graduate population.

In general, this project scope paves the way to a realistic and manageable implementation as it targets ICT TVET graduates in Kicukiro District, involves a limited and appropriate sample size, and provides a technically feasible system that will be subject to pilot testing and expansion in the future.

1.6 Significance and Justification

It is hoped that the successful application of the Skills Gap Analysis Tool will enhance ICT TVET graduates' competency awareness compared to the National Occupational Standards. Structured self-assessment and clear feedback enable learners to have a better sense of their strengths and weaknesses, and develop skills more effectively.

The system will also help training institutions and the Rwanda TVET Board (RTB) to get evidence-based data on the prevailing gaps in skills of ICT learners in Kicukiro district. This can help to enhance the delivery of curriculum, link curriculum to industry requirements, and improve the outcomes of competency-based education.

Moreover, it is expected that the tool will help to improve the employability of graduates by closing the gap between training and expectations of the labour market. Employers will have better job seekers, graduates will be more aware of recruitment processes, thereby reducing unemployment and a better skilled ICT labour force in Rwanda.

1.7 Research Budget

This section presents the estimated cost needed to effectively put the research project into practice on a Skills Gap Analysis Tool for ICT TVET Graduates in Kicukiro District. The budget only covers costs that are necessary and feasible, including internet, development tools, travel, and other costs during the data collection, system design and implementation phases.

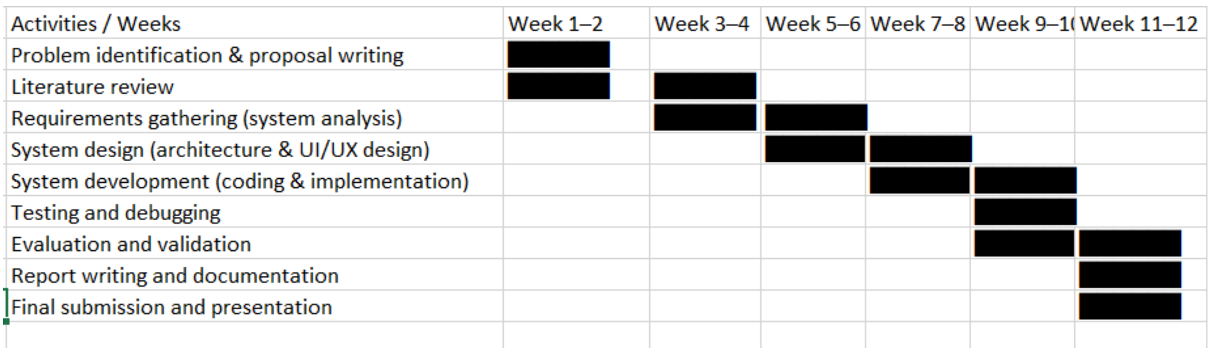
No	Item Description	Justification	Unit Cost (RWF)	Quantity	Total Cost (RWF)
1	Internet bundles	Used for research, online documentation, and system development	10,000	3 months	30,000
2	Transport costs	Field visits to ICT TVET institutions in Kicukiro District for data collection	5,000	12 trips	60,000
3	Printing & photocopying	Printing questionnaires, proposal documents, and final report	200	150 pages	30,000
4	Stationery (notebooks, pens)	Used for note-taking during interviews and research activities	10,000	1 set	10,000
5	Cloud hosting/domain (basic)	Hosting prototype system or testing environment online	40,000	1	40,000
6	Software tools (design & development support)	Tools such as UI design or development support platforms (e.g., premium plugins if needed)	20,000	1	20,000

7	Contingency fund	Unexpected expenses during research activities	-	-	15,000
Total Estimated Cost					205,000 RWF

1.8 Research Timeline

The implementation of this research project will be carried out over a period of three months, with activities organised sequentially to ensure systematic development, testing, and evaluation of the proposed system. The Gantt chart below illustrates the timeline of key project phases and their corresponding duration.

1.8.1 Gantt Chart of Activities



1.8.2 Explanation of the Timeline

Problem identification and proposal development are the preliminary steps in the process of the research. This is followed by a literature review that aims to get an understanding of the existing literature, skills gap analysis systems, and competency frameworks on TVET.

In the 2nd phase of the process, requirements will be gathered through an analysis of the needs of the users, such as ICT TVET graduates in Kicukiro district. This ensures that the system design is in relation to the needs in the real situation.

The system design phase includes the design of the Skills Gap Analysis Tool's architecture and database design, as well as the design of the user interface. Once the design is complete, the project enters the system development phase where the software is actually designed.

After development, the system is tested and debugged to check for and correct errors, making it work reliably. Evaluation and validation take place after this, where the system is evaluated by comparing competencies to RTB occupational standards.

Finally, the project ends with report writing, documentation, final submission and presentation that summarises the research outcomes, implementation and the outcomes of the project.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This review was limited to software-related research and digital tools on competency and skills gaps assessment in TVET, especially ICT. A systematic literature search of scholarly databases IEEE Xplore, ACM Digital Library, ScienceDirect, and Google Scholar, as well as official repositories (UNESCO-UNEVOC, ILO, Rwanda TVET Board, RISA), was carried out. The keywords used were: TVET competency assessment, skills gap analysis software, digital learning management Rwanda, and similar terms. More than 30 relevant studies, reports and policy documents were initially identified, out of which 14 are deemed to be highly relevant for in-depth discussion. The sources selected were based on their academic rigour and focus on the direct relevance to software systems or digital systems for competency assessment as applied to the Rwanda context and similar contexts in TVET.

2.2 Existing Competency Assessment Platforms

A variety of digital platforms exist for mapping competencies and analysing skill gaps, though most target corporate or general education contexts. Learning Management Systems (LMS) like Moodle Workplace or iSpring Learn include built-in competency frameworks and gap-reporting features (e.g. 360° skill surveys and analytics). Talent-management systems (e.g. TalentGuard, Mercer) and specialised software (e.g. Lepaya) employ AI-driven skills intelligence for workforce planning. These systems demonstrate that automated competency tracking and skills-gap reporting are technically feasible. However, most of these platforms were designed for corporate workforce management rather than competency-based TVET education environments. As a result, they lack alignment with Rwanda TVET Board occupational standards and do not adequately support learner-centred competency assessment within developing-country educational contexts.

For example, international studies note that companies increasingly use LMS, AI chatbots and analytics dashboards to continuously assess employee skills and personalise training (Braun, n.d.). However, these corporate tools assume predefined competency models and often rely on self-assessment or supervisor input, which can suffer from inconsistent usage and accuracy issues (Braun, n.d.).

International TVET initiatives have produced self-assessment "toolkits" for providers (e.g. on digitalisation, inclusion, quality assurance) that allow institutions to reflect on gaps in their curricula and management (*Toolkits*, 2017). UNESCO-UNEVOC's online repository, for instance, catalogues free toolkits (checklists, surveys, guidelines) for TVET quality and digital readiness (*Toolkits*, 2017). These, however, are institutional tools (for administrators) rather than automated learner-centred platforms. We found no existing software solution specifically designed to match TVET learners' assessed competencies against national occupational standards or to give personalised skill-gap feedback. In Rwanda, current digitalisation efforts (e.g. the LuxDev "Digital Skills for Quality TVET" project) focus on strengthening ICT infrastructure, online/blended teaching, and data systems, but do not provide automated individual skill-gap diagnostics (*Digital Skills for Quality TVET in Rwanda*, 2025). In sum, while many generic LMS and HR systems support competency tracking, none are tailored to Rwanda's TVET curriculum or RTB occupational standards, indicating a gap that our proposed tool aims to fill.

2.2.1 Novelty Comparison of Existing Platforms and Proposed System

Literature reviewed shows competencies management and skills gaps analysis systems are already in place in educational and corporate settings. Competency tracking, training management and workforce development features are offered by platforms like Moodle Workplace, iSpring Learn, TalentGuard and Lepaya. These platforms are targeted mainly for general education and for corporate training and development, and are not directly relevant to Rwanda TVET Board (RTB) occupational standards or to the needs of ICT TVET graduates in Rwanda.

Table 2.1: Novelty Comparison of Existing Platforms and Proposed System

Platform	Competency Tracking	Skills Gap Analysis	Personalised Recommendations	RTB Occupational Standards Integration	Rwanda TVET Context Support	Primary Target Users
Moodle Workplace	Yes	Limited	Limited	No	No	Educational institutions and organisations

iSpring Learn	Yes	Limited	Limited	No	No	Corporate training environments
TalentGuard	Yes	Yes	Yes	No	No	Corporate workforce management
Lepaya	Yes	Partial	Yes	No	No	Employee upskilling and workforce development
Proposed Skills Gap Analysis Tool	Yes	Yes	Yes	Yes	Yes	ICT TVET graduates and TVET institutions

The comparison shows that competency mapping, skills-gap dashboards and recommendation systems are already commonplace functions in the existing platforms. However, none of the systems reviewed includes Rwanda TVET Board occupational standards in the assessment process or offers a competency-gap analysis tailored to the context of Rwanda for the ICT TVET graduates.

The novelty of the proposed system is not the competency tracking per se, but the ability to map graduate competencies to RTB occupational standards, identify gaps in competencies in relation to the standards, and provide personalised recommendations based on the Rwanda competency-based TVET framework. The main contribution of the proposed system is the integration of RTB standards and alignment to the Rwanda context, a gap identified in the literature.

2.3 Review of Related Work

2.3.1 Skills Gap Challenges in TVET Education

Numerous studies document persistent gaps between the skills TVET graduates possess and those demanded by industry in Rwanda and similar contexts. For example, Diop (2020) observes that “Rwanda is currently experiencing a huge skills gap, particularly digital skills, resulting in critical economic development barriers” (Mugiraneza, n.d.)

. She notes that over half of TVET students train only in traditional trades (construction, mechanics) rather than ICT or modern technologies (Mugiraneza, n.d.)

. Similarly, the Rwanda State of Skills Supply and Demand report (2024) found that skills mismatches contribute to above-average youth unemployment. It reports that graduates in business, tourism and humanities are “overproduced vis-à-vis” high-demand fields like engineering and ICT (*Rwanda State of Skills Supply and Demand Report 2024*, 2024), implying shortages in technical fields. Other researchers confirm that many Rwandan TVET institutions still use outdated training methods and lack alignment with digital workplace requirements. For instance, a 2022 study reported UNESCO findings that TVET schools suffer “shortage of equipment and infrastructures … affect[ing] acquisition of technical skills” (Hakizayezu & Maniraho, 2022), and that many graduates lack basic prerequisite or ICT skills needed for entry-level jobs (Hakizayezu & Maniraho, 2022). In sum, the literature consistently highlights that, despite Rwanda’s education reforms, most TVET graduates do not fully meet employers' ICT skill needs due to gaps in curriculum, teaching, and assessment methods (Mugiraneza, n.d.).

(*Rwanda State of Skills Supply and Demand Report 2024*, 2024) (Hakizayezu & Maniraho, 2022).

2.3.2 Competency-Based Education and Occupational Standards

Rwanda has formally adopted Competency-Based Education and Training (CBET) to address these skill gaps. The Rwanda TVET Board (RTB) and related agencies have developed national occupational standards to guide curriculum and assessments. A strategic RTB plan emphasises CBET as key to workforce readiness (Niyonasenze et al., 2024). Likewise, Rwanda’s National Digital Skills Framework (2023) defines seven digital competency areas (e.g. information literacy, digital communication, problem-solving) and four proficiency levels to align education with labour market needs (*National Digital Skills Framework*, 2023). The NDSF explicitly aims to “equip graduates with the skills required by industry” by aligning teaching outcomes with digital economy demands (*National Digital Skills Framework*, 2023). These frameworks provide strong foundations for TVET modernisation. However, several authors note weak implementation: for example, Niyonasenze et al. (2024)

conclude that despite Rwanda's commitment, "the country still faces serious gaps in terms of technically skilled and a trained labour force" (Niyonasenze et al., 2024). Mugiraneza (ILO 2021) similarly found that only half of schools surveyed had access to digital learning content, and many teachers lack ICT skills, limiting effective CBET delivery (Mugiraneza, n.d)

. Thus, while Rwanda has high-level competency standards, the literature suggests that monitoring and assessment systems are insufficient. There is a consensus that TVET institutions lack learner-centred software tools to operationalise these standards – i.e. to assess individual students against the RTB-defined competencies and generate personalised feedback (Niyonasenze et al., 2024) (Mugiraneza, n.d)

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2.3.3 Digitalisation and Skills Assessment Technologies

Digital technologies hold promise for closing skill gaps by enabling continuous, data-driven assessment and personalised learning. Global reviews note the rise of "intelligent" learning systems that use automation and analytics to track competencies and recommend training (Braun, n.d.). However, the adoption of such systems in education has been limited. Braun *et al.* (2025) found that companies using LMS, AI chatbots and scenario planning still struggle with immature integration and unreliable self-assessments (Braun, n.d.). Education researchers similarly call for integrating digital skills across curricula. Zhong and Juwaheer (2024) argue for a "whole-institution" approach in TVET, where leaders, teachers and students all develop digital competencies in tandem (Zhong & Juwaheer, 2024). They emphasise that TVET's commitment to digital competency must address skill gaps at all stakeholder levels (Zhong & Juwaheer, 2024).

In Rwanda, national initiatives recognise this need: the LuxDev project "Digital Skills for Quality TVET" (2024–28) aims to modernise Rwanda's TVET by improving ICT infrastructure, training teachers in digital pedagogy, and strengthening data systems for monitoring (*Digital Skills for Quality TVET in Rwanda*, 2025). Yet these efforts focus on institutional capacity rather than on individual assessments. International analyses warn that without targeted tools, even advanced policies leave skill gaps unaddressed. For example, a recent Brookings study notes that "there is limited research aimed at measuring and understanding ... digital skills gaps in Africa" (Bhorat et al., 2023), underscoring that digital skills-gap analysis remains underdeveloped. It also highlights the urgency for Africa's workforce: as the world's youngest continent, Africa will comprise one-fifth of the global workforce (and one-third of the youth workforce) by 2030 (Bhorat et al., 2023). This demographic pressure makes effective skills assessment and development critical. In sum, while digital tools for competency tracking exist globally (Braun, n.d.), in Rwanda they have yet to be integrated into TVET. The literature points to a significant opportunity: deploying an automated digital assessment system aligned with RTB standards could provide the continuous feedback and personalised learning needed to bridge Rwanda's ICT skills gap (*Digital Skills for Quality TVET in Rwanda*, 2025) (Zhong & Juwaheer, 2024).

2.4 Summary of Reviewed Literature

The key findings from the literature are:

A. ·Persistent Skills Mismatches: Multiple sources confirm significant gaps between TVET graduate skills and labour market needs in Rwanda, especially in ICT and modern technical fields (*Rwanda State of Skills Supply and Demand Report 2024*, 2024) (Hakizayezu & Maniraho, 2022).
B. Strong Policy Frameworks: Rwanda has well-defined competency frameworks (RTB occupational standards, NDSF digital skills areas) intended to align training with industry requirements (*National Digital Skills Framework*, 2023) (Niyonasenze et al., 2024).

C. ·Digital Assessment Feasibility: Existing international LMS and corporate platforms demonstrate that automated skills-gap analysis (via competency tracking, 360° assessments, analytics dashboards) is technically possible (Braun, n.d.).

D. ·Reliance on Traditional Assessments: Despite policies, many Rwandan TVET programs still rely on exams and ad hoc evaluations rather than continuous digital assessment. Graduates often complete TVET without formal self-assessment tools or feedback loops (Hakizayezu & Maniraho, 2022) (Niyonasenze et al., 2024).

E. Generic vs. Local Systems: Most available LMS and talent-management tools are generic and designed for business settings. None are customised for Rwanda's TVET curricula or RTB standards. There is currently no existing software that lets Rwandan ICT students assess their skills directly against occupational standards and generate personalised learning recommendations.

These points collectively highlight a clear gap: the absence of a purpose-built digital platform for competency assessment in Rwandan TVET, which justifies the development of the proposed Skills Gap Analysis Tool.

2.5 Strengths and Weaknesses of Existing Systems

Strengths:

A. Rwanda's RTB competencies and national digital skills framework are a strong starting point for any assessment tool (*National Digital Skills Framework*, 2023). These policies give clear definitions of desired skills across sectors.

B. International LMS and HR systems illustrate that automated competency mapping, skills-gap visualisation and tailored training are feasible in software (Braun, n.d.). Technical components such as 360-degree evaluations and reporting dashboards have been successfully implemented in other contexts.

C. UNESCO/ILO guidance and toolkits provide proven best practices for quality assurance and digitalisation in TVET (e.g. self-reflection toolkits) (*Toolkits*, 2017). These resources, while not software, offer models for content and assessment criteria.

D. Ongoing digitalisation initiatives (such as the LuxDev project) are improving ICT infrastructure, connectivity and data systems in TVET schools (*Digital Skills for Quality TVET in Rwanda*, 2025). This growing capacity will support any new digital platform deployment.

Weaknesses:

A. Existing LMS/workforce systems are not aligned with Rwanda's specific occupational standards. They typically support general competency categories, not the detailed RTB-defined skills.

B. Most available platforms emphasise organisational training management (e.g. tracking employee training in companies) rather than individual student self-assessment. They lack features for student-centred, CBET-style skills evaluation.
C. Current TVET assessments in Rwanda focus on summative exams and teacher evaluations, without structured self-assessment or continuous diagnostics. There is no centralised repository linking student performance data to the national curriculum standards.
D. No existing software provides automated, individualised feedback on how to close identified skill gaps. In other words, available systems do not suggest personalised learning paths based on a student's competency profile.

These weaknesses point to the need for a dedicated solution: a tool that integrates Rwanda's RTB standards with automated competency testing and gap analysis for ICT TVET learners.

2.6 Conclusion

The literature makes clear that Rwanda continues to face ICT-related skill shortages despite strong CBET policies and increasing digital investments. National frameworks define the competencies graduates should have (*National Digital Skills Framework*, 2023), and global research shows automated skills-gap analysis is technically feasible (Braun, n.d.). Yet in practice, Rwandan TVET systems lack continuous digital assessment tools. Existing platforms are generic and aimed at corporate or institutional use (Braun, n.d.) (*Toolkits*, 2017), rather than enabling students to self-evaluate against RTB standards. Consequently, no tool currently exists to automatically compare ICT students' abilities with Rwanda's occupational requirements and offer targeted advice.

This review thus identifies a clear technological and research gap. Developing a specialised Skills Gap Analysis Tool – one that operationalises RTB standards and the NDSF through online assessments and personalised feedback – could address several needs at once. Such a system would empower learners to recognise their strengths and weaknesses relative to industry expectations, aid instructors in tracking competency progress, and help institutions align curricula with market demand (*National Digital Skills Framework*, 2023) (*Digital Skills for Quality TVET in Rwanda*, 2025). In the context of Africa's digital transformation and

Rwanda's youth workforce surge (Bhorat et al., 2023), this solution has the potential to improve graduate readiness and employability.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the system analysis and design for the proposed Skills Gap Analysis Tool for ICT TVET Graduates in Kicukiro District, Rwanda. The chapter explains the methodologies and approaches used in designing the system, including the research methodology, development model, system architecture, UML diagrams, database structure, and development tools.

The purpose of the proposed system is to analyse and assess the ICT competencies possessed by TVET graduates against the Rwanda TVET Board (RTB) occupational competency standards. The system is intended to help ICT TVET graduates identify their competency strengths and weaknesses through automated skills-gap analysis and personalised feedback.

The proposed system will mainly support two categories of users: ICT TVET graduates and system administrators or institutional managers responsible for managing occupational standards and competency requirements. Graduates will use the system to assess their ICT skills, while administrators will manage competency standards, assessment criteria, and reports.

The system design process focuses on creating a scalable, user-friendly, secure, and efficient web-based platform capable of collecting, processing, and analysing graduate competency assessment data. The research adopted a software development methodology that supports iterative development, flexibility, continuous improvement, and user feedback throughout implementation.

3.2 Research Design (including the development model used)

3.2.1 Research Methodology

This study will adopt a mixed-methods research approach that combines both qualitative and quantitative methods of data collection and analysis.

The qualitative approach will help in understanding the challenges faced by ICT TVET graduates regarding competency assessment, skills development, and alignment with RTB occupational standards. The quantitative approach will support the collection of measurable data related to competency gaps, system usability, assessment accuracy, and user satisfaction.

The mixed-methods approach is appropriate for this study because it provides comprehensive insights into both the technical and educational aspects associated with competency-based assessment systems.

Data will be collected using questionnaires, interviews, and document analysis from selected ICT TVET graduates and institutional administrators within Kicukiro District.

3.2.2 Sampling Strategy

The study will use purposive sampling to select participants who are directly relevant to the research objectives.

1. The target population will include:
2. ICT TVET graduates
3. TVET instructors
4. Institutional administrators responsible for competency management

A sample size of approximately 50–120 ICT TVET graduates will be selected for system testing and evaluation. In addition, a small number of institutional administrators and instructors will participate in interviews and feedback sessions to provide insights regarding ICT competency standards and assessment requirements.

Purposive sampling is suitable for this study because it allows the collection of information from participants with relevant experience and knowledge related to ICT competency assessment and competency-based education.

3.2.3 Development Model

The development of the Skills Gap Analysis Tool follows the Agile Software Development Model.

Agile methodology was selected because it allows continuous improvement, flexibility, stakeholder involvement, and iterative development throughout the project lifecycle.

The Agile model is suitable for this project because system requirements may evolve based on feedback from ICT TVET graduates and institutional administrators. The iterative nature of Agile enables continuous testing, refinement, and improvement of system functionalities.

3.2.4 Agile Development Phases

Requirement Gathering

1. Collect information from ICT TVET graduates and institutional administrators
2. Analyse RTB occupational competency standards
3. Identify system functional and non-functional requirements

System Planning

1. Define system objectives and project scope
2. Select technologies and development resources
3. Plan implementation activities and timelines

System Design

1. Design database structure
2. Design system workflows and interfaces
3. Develop UML diagrams and system architecture

Development

1. Implement frontend interfaces
2. Develop backend APIs and business logic
3. Implement skills-gap analysis algorithm
4. Integrate database functionalities

Testing

1. Perform unit testing
2. Conduct integration testing
3. Perform user acceptance testing
4. Validate competency analysis accuracy

Deployment

1. Deploy the system to a web hosting environment
2. Configure server and database settings

Maintenance

1. Improve the system based on user feedback
2. Update competency standards when necessary
3. Fix bugs and improve system performance

3.2.5 Competency Measurement and Gap Scoring Mechanism

The core objective of the proposed Skills Gap Analysis Tool is to evaluate the competency of ICT TVET graduates against Rwanda TVET Board (RTB) competency standards of the occupational skills and competencies and determine competency gaps that could hinder employability.

The competency assessment process includes four steps: collecting competency evidence, scoring competency evidence, mapping competency evidence to RTB standards, and conducting a gap analysis.

Stage 1: Competency Evidence Collection

To ensure that competency assessment is evidence-based rather than purely self-reported, the system will collect competency evidence from multiple sources.

Table 3.1: Competency Evidence Sources

Evidence Source	Description	Weight (%)
Practical Assessment	Competency-based practical tasks aligned with RTB occupational standards	40
Portfolio Evidence	Projects, assignments, GitHub repositories, and completed practical work	30
Academic Records	Institutional competency assessment records and academic achievements	20
Self-Assessment	Graduate self-evaluation of competency confidence	10
Total		100

Several sources of evidence enhance the reliability of assessment as well as limit the reliance on self-report.

Stage 2: Competency Scoring

For each competency area, the system calculates a weighted competency score using the collected evidence.

Competency Score = (Practical Assessment × 0.40) + (Portfolio Evidence × 0.30) + (Academic Records × 0.20) + (Self-Assessment × 0.10)

The resulting score represents the graduate's demonstrated competency in a specific ICT competency area.

Stage 3: Mapping to RTB Occupational Standards

Each assessed competency is mapped to the corresponding competency defined in the RTB occupational standards.

For example:

Graduate Competency	RTB Occupational Competency
SQL Database Design	Database Management
Website Development	Web Development
Network Configuration	Computer Networking

The weighted competency score is then converted into a competency level.

Table 3.2: Competency Level Classification

Competency Score (%)	Competency Level
80–100	Level 4 – Highly Competent
60–79	Level 3 – Competent
40–59	Level 2 – Partially Competent
0–39	Level 1 – Not Yet Competent

The competency level to be achieved in each competency area will be derived from RTB occupational standards and confirmed by reference to ICT instructors and competency assessors.

Stage 4: Gap Analysis and Recommendation Generation

The system compares the graduate's competency level with the required RTB competency level.

Gap Score = Required Competency Level − Graduate Competency Level

The gap score determines the severity of the competency gap.

Table 3.3: Gap Classification Framework

Gap Score	Classification	Recommendation Type
0	No Gap	Maintain competency level
1	Low Gap	Minor competency improvement
2	Moderate Gap	Additional practical training required
3 or Above	High Gap	Intensive upskilling required

The system creates personalised recommendations (suggested training areas, practical exercises and competence development activities) based on the identified gap, which are necessary to reach the RTB standard.

3.2.6 Worked Example of Competency Assessment and Gap Calculation

The example shows how competency evidence can be translated to a competency score as per RTB standards, and how a skills-gap report can be generated.

Suppose a graduate submits evidence for the competency area "Database Management."

The system evaluates the evidence as follows:

Table 3.4: Competency Assessment Results

Evidence Source	**Score (%)**	**Weight (%)**	**Weighted Score**
Practical Assessment	80	40	32
Portfolio Evidence	70	30	21
Academic Records	75	20	15
Self-Assessment	90	10	9
Final Competency			77%

Score			

The final competency score is calculated as:

Competency Score = 32 + 21 + 15 + 9

Competency Score = 77%

According to Table 3.2, a score of 77% corresponds to:

Graduate Competency Level = Level 3 (Competent)

The RTB occupational standard for Database Management requires:

Required Competency Level = Level 4 (Highly Competent)

The gap score is therefore calculated as:

Gap Score = 4 − 3

Gap Score = 1

Table 3.5: Gap Analysis Result

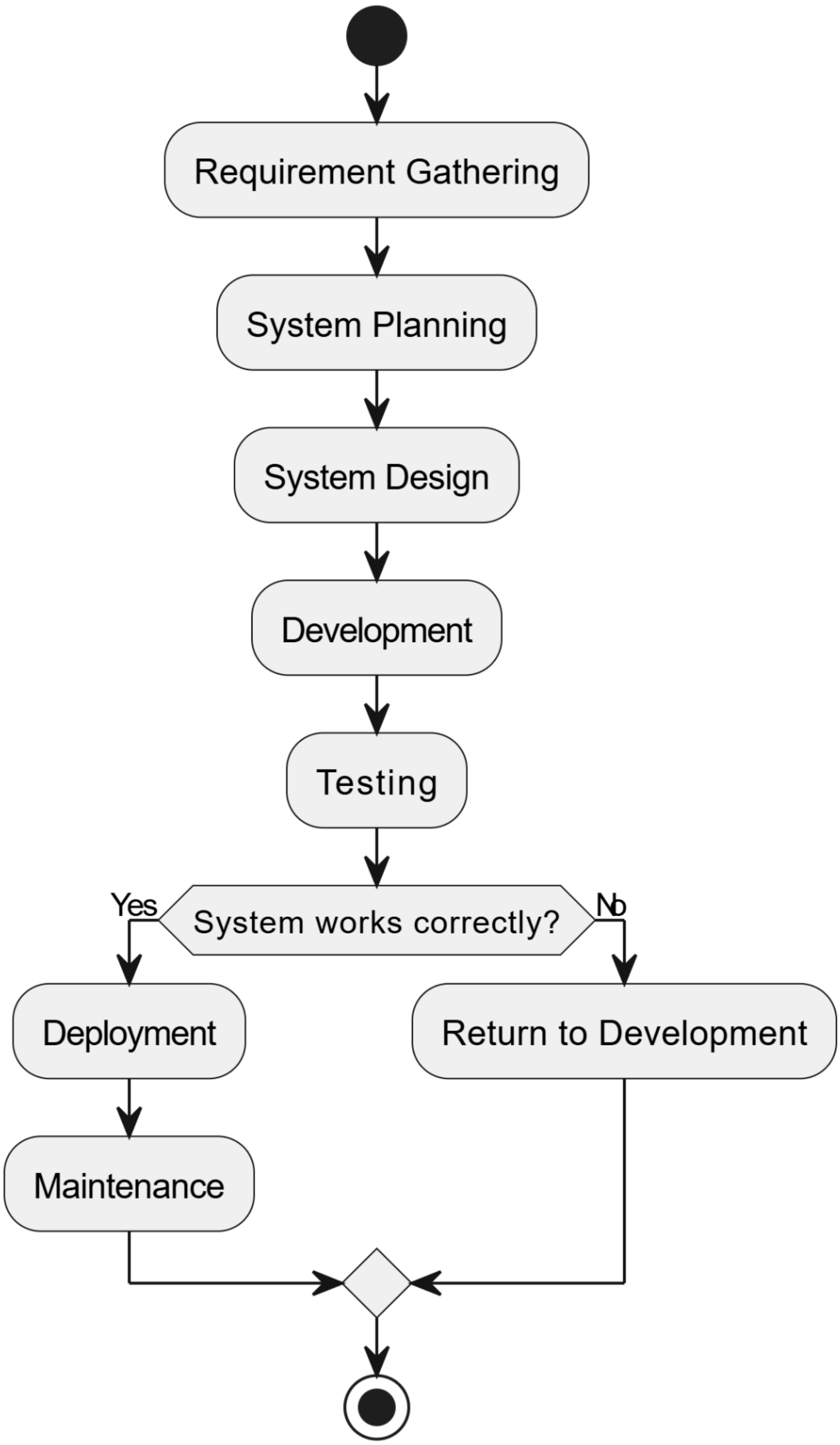
Competency	Required Level	Graduate Level	Gap Score	Classification
Database Management	Level 4	Level 3	1	Low Gap

Given this outcome, the system makes the following suggestions:

Develop proficiency in more complex and advanced database management skills by means of further SQL optimisation exercises, further practice in database administration and project-based learning.

This example covers the entire workflow of the proposed system, from the collection of competency evidence to identification of gaps and generation of recommendations.

3.2.7 Proposed Development Model Diagram


Requirement Gathering
System Planning
System Design
Development
Testing
System works correctly?
Yes
No
Deployment
Return to Development
Maintenance

3.2.8 Functional Requirements

The proposed system shall:

1. Allow graduates to register and log into the system
2. Allow graduates to complete practical assessments and upload portfolio evidence for competency evaluation.
3. Allow administrators to manage RTB occupational competency standards
4. Compare graduate competencies against RTB ICT standards
5. Perform automated skills-gap analysis
6. Generate competency assessment reports
7. Provide recommendations for skills improvement
8. Allow administrators to manage users and assessment records
9. Store graduate assessment history
10. Display competency scores and analysis dashboards

3.2.9 Non-Functional Requirements

The system should:

1. Provide secure authentication and data protection
2. Maintain confidentiality of graduate assessment data
3. Support concurrent users efficiently
4. Be accessible through modern web browsers
5. Maintain fast response time during competency analysis
6. Provide a user-friendly interface
7. Ensure scalability for future expansion
8. Provide system reliability and availability
9. Support responsive design for desktop and mobile devices

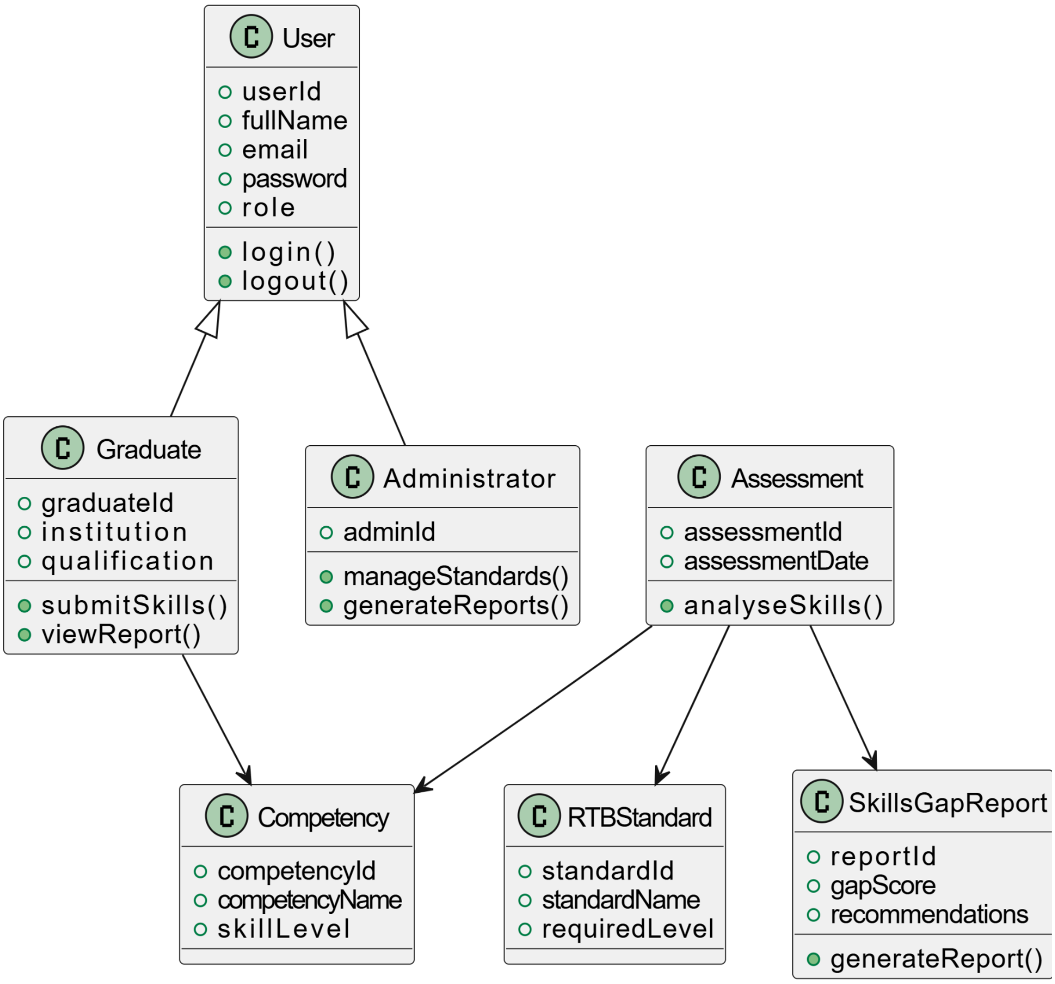
3.3 Class Diagram

The class diagram represents the static structure of the proposed Skills Gap Analysis Tool. It illustrates the relationships between major system classes such as Users, Graduates, Competencies, Standards, Assessments, and Reports.

The diagram helps define system objects, attributes, methods, and interactions necessary for implementing the application.

Main Classes

1. User
2. Graduate
3. Administrator
4. Competency
5. RTBStandard
6. Assessment
7. SkillsGapReport



3.4 System Architecture

The proposed system uses a three-tier full-stack web architecture consisting of:

1. Presentation Layer (Frontend)
2. Application Layer (Backend/API)
3. Database Layer

This architecture was selected because it supports scalability, maintainability, security, and efficient communication between system components.

Architecture Components

1. Presentation Layer (Frontend)

This layer provides the user interface accessible through web browsers. Graduates and administrators interact with the system through responsive web pages.

Technologies

1. HTML5
2. CSS3
3. JavaScript
4. React.js

2. Application Layer (Backend/API)

This layer processes business logic, authentication, skills-gap analysis, report generation, and communication between the frontend and the database.

Technologies

1. Node.js
2. Express.js
3. RESTful APIs

3. Database Layer

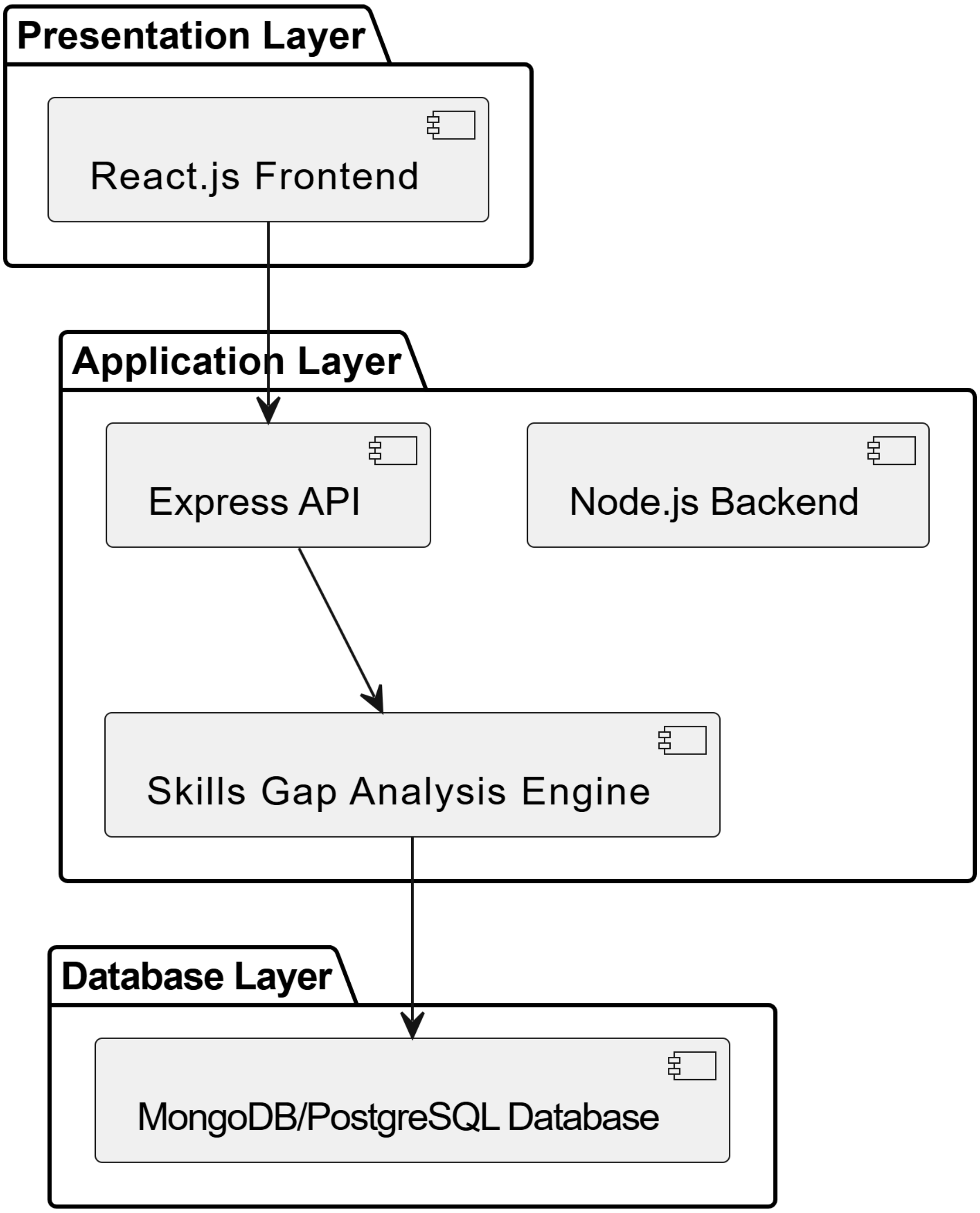
This layer stores system data including:

1. User profiles
2. ICT competencies
3. RTB occupational standards
4. Assessments
5. Skills-gap reports

Technology

1. MongoDB or PostgreSQL

Full Stack System Architecture Diagram



3.5 UML Diagrams

Unified Modelling Language (UML) diagrams are used to visually represent system processes, interactions, workflows, and data relationships.

The following UML diagrams were developed for the proposed system:

1. Use Case Diagram
2. Activity Diagram
3. Sequence Diagram
4. Entity Relationship Diagram (ERD)

3.5.1 Use Case Diagram

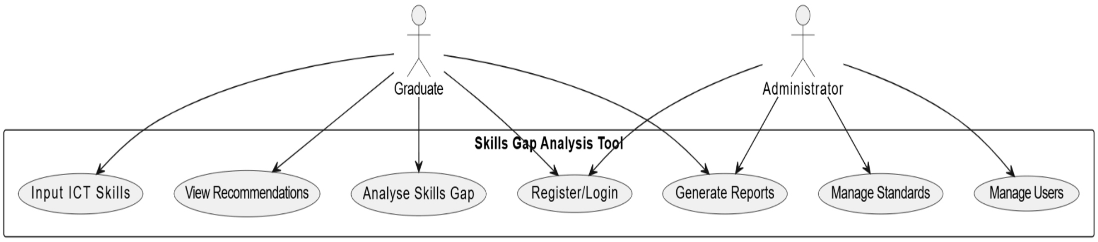
The use case diagram illustrates how users interact with the system.

Main Actors

1. Graduate
2. Administrator

Main Functionalities

1. Register/Login
2. Input ICT Skills
3. Analyse Skills Gap
4. Generate Assessment Reports
5. View Recommendations
6. Manage Competency Standards
7. Manage Users

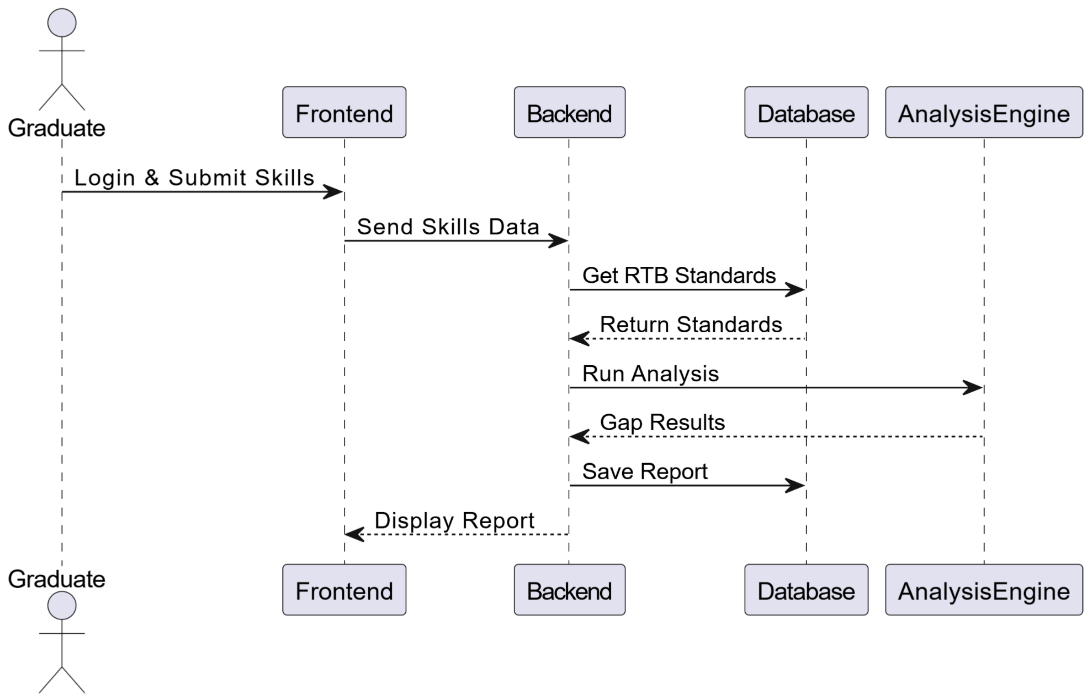


3.5.2 Activity Diagram

The activity diagram shows the workflow of the skills-gap analysis process.

3.5.3 Sequence Diagram

The sequence diagram shows interactions between system components over time.

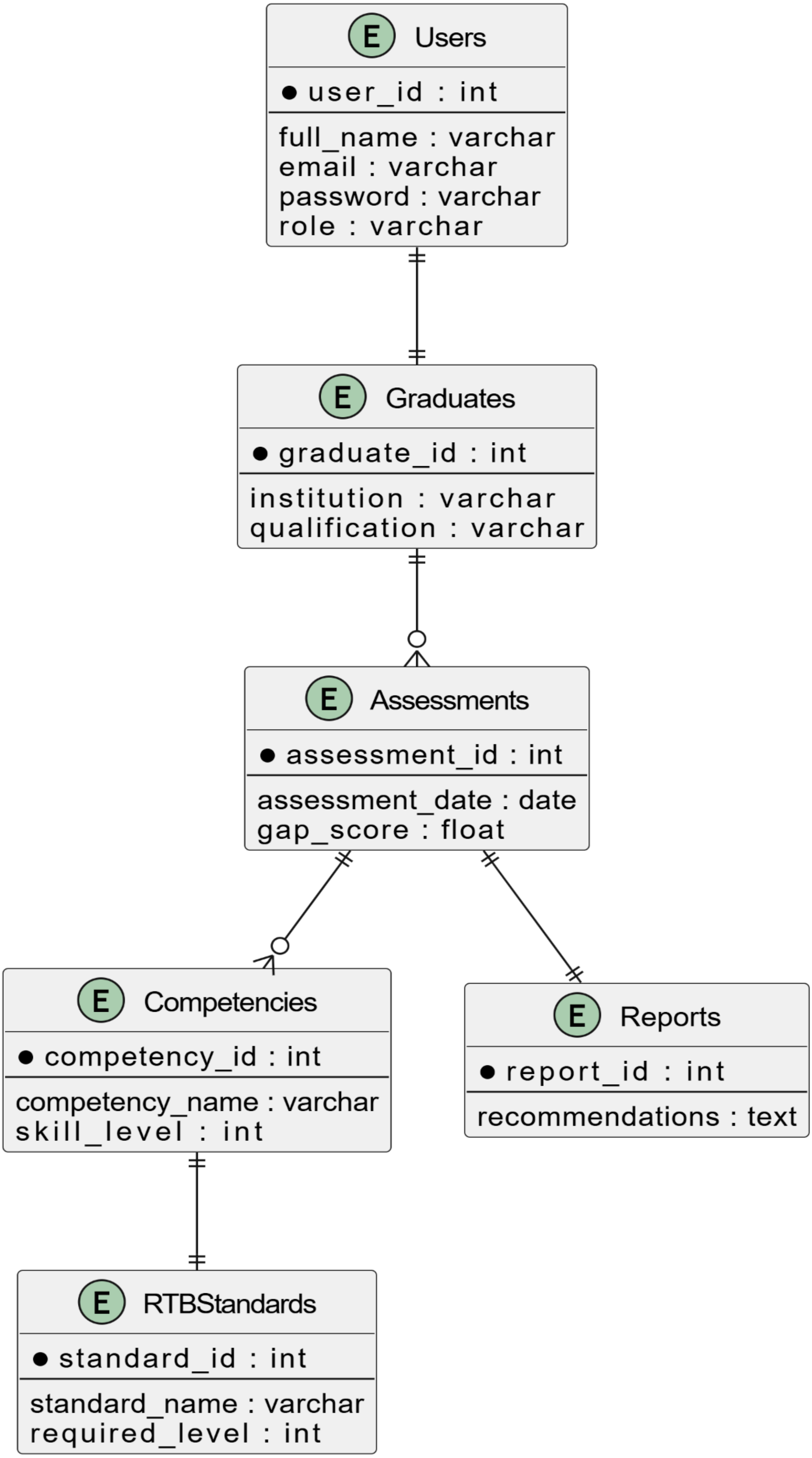


3.5.4 Entity Relationship Diagram (ERD)

The ERD illustrates relationships among database entities.

Main Entities

1. Users
2. Graduates
3. Competencies
4. RTBStandards
5. Assessments
6. Reports



E Users
user_id : int
full_name : varchar
email : varchar
password : varchar
role : varchar
E Graduates
graduate_id : int
institution : varchar
qualification : varchar
E Assessments
assessment_id : int
assessment_date : date
gap_score : float
E Competencies
competency_id : int
competency_name : varchar
skill_level : int
E Reports
report_id : int
recommendations : text
E RTBStandards
standard_id : int
standard_name : varchar
required_level : int

3.5.5 Ethical Considerations

This research will follow ethical research principles throughout the data collection, system development, and evaluation processes.

Participants involved in the study will be informed about the purpose of the research before participating. Participation will be voluntary, and respondents will have the right to withdraw from the study at any time without negative consequences.

Informed consent will be obtained from all participants before collecting data.

The confidentiality and privacy of participants will be maintained by ensuring that personal information is not disclosed to unauthorised individuals. Data collected during the study will be used strictly for academic purposes and securely stored to prevent unauthorised access.

The system will also implement secure authentication mechanisms and encrypted password storage to enhance data protection and privacy.

In addition, this will ensure academic honesty by properly acknowledging all sources of information using APA 7th edition referencing guidelines.

3.5.6 System Evaluation Design

The proposed Skills Gap Analysis Tool will be assessed by a comprehensive assessment framework designed to meet the study objectives. The evaluation is designed to find out if the system is able to measure competencies of the graduates of ICT TVET, if it provides accurate skills gap analysis, if it provides useful skills gap analysis recommendations, and if it raises awareness of the graduates' competencies in ICT.

ICT TVET graduates, ICT TVET instructors and TVET institutional administrators from selected TVETs in Kicukiro District will be involved in the evaluation. Data will be gathered using pre and post-assessment surveys, expert review, and system-generated assessment records, as well as questionnaires on usability.

Table 3.6: Evaluation Design Framework

Research Objective	Evaluation Indicator	Metric	Data Source	Evaluation Method
Assess the competency levels of at least 50 ICT TVET graduates across selected RTB occupational competencies	Number of graduates successfully assessed and competency reports generated	Number of competency assessments completed and competency scores produced	System assessment records	Descriptive statistical analysis
Design and develop a web-based Skills Gap Analysis Tool that maps competencies against RTB standards and generates recommendations	Successful implementation of core system functionalities	Percentage of functional requirements successfully implemented	Functional testing reports and system test cases	Functional testing
Test and evaluate the effectiveness, usability, and accuracy of the developed system	System effectiveness, usability, and assessment accuracy	User satisfaction score (≥4/5), expert agreement rate (≥80%), successful task completion rate	User questionnaires, expert validation reports, and system logs	Usability testing and expert evaluation
Evaluate whether the system improves competency awareness among ICT TVET graduates	Change in competency-awareness levels before and after system use	Difference between pre-assessment and post-assessment competency-awareness scores	Pre-assessment and post-assessment survey responses	Pre-test and post-test comparison

Evaluation Success Criteria

The proposed system will be considered successful if:

1. At least 50 ICT TVET graduates complete competency assessments through the system.
2. All core functional requirements are successfully implemented and tested.
3. The system achieves at least 80% agreement between generated gap reports and expert assessment results.
4. The system achieves an average usability rating of at least 4 out of 5 from users.
5. Graduates demonstrate a measurable improvement in competency-awareness scores after using the system.
6. The system successfully generates competency-gap reports and personalised recommendations aligned with RTB occupational standards.

The results obtained from the evaluation process will be used to determine whether the proposed Skills Gap Analysis Tool effectively supports competency assessment, skills-gap identification, and competency awareness improvement among ICT TVET graduates in Kicukiro District.

3.6 Development Tools

The following tools and technologies will be used during the development of the Skills Gap Analysis Tool.

Tool/Technology	Purpose
HTML5	Structuring web pages
CSS3	Styling user interfaces
JavaScript	Client-side scripting
React.js	Frontend framework
Node.js	Backend runtime environment
Express.js	Server-side framework
MongoDB/PostgreSQL	Database management
Visual Studio Code	Source code editor
Git & GitHub	Version control and collaboration
Draw.io	UML and system diagram creation
Postman	API testing
Figma	User interface design
RESTful API	To enable communication between frontend and backend systems through HTTP requests

3.7 Conclusion

This chapter presented the system analysis and design of the proposed Skills Gap Analysis Tool for TVET Graduates in ICT – Kicukiro District. The chapter discussed the Agile development model, class diagram, system architecture, UML diagrams, ERD, and development tools. The proposed design provides a strong foundation for implementing a scalable, secure, and efficient web-based system capable of analysing skills gaps between ICT graduates and labour market demands.

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